

**University of Bedfordshire
Faculty of Creative Arts, Technologies and Science
School of Computer Science and Technology**

Comparative Integrated Systems

MAIN WITH ANSWERS

Date of Exam: Assessment Department to complete

Start Time: Assessment Department to complete

Time Allowed: 2 Hours

Unit Co-ordinator: Marc CONRAD

INSTRUCTIONS TO CANDIDATES

Answer ALL Questions.

- Total 100 marks.
- This examination is a closed book paper.
- No additional materials are allowed in the examination room.
- Write your answers in the answer book provided.
- Use of a Calculator is NOT allowed in the exam.

**PLEASE DO NOT OPEN THIS PAPER UNTIL INSTRUCTED TO DO SO BY
THE SENIOR INVIGILATOR**

Comparative Integrated Systems

(main paper)

Answer ALL Questions

Question 1 [10 marks]

The CORBA standard is defined by the Object Management Group (OMG); the language XML is defined by the W3C Consortium. Who are typically the members of organisations such as the OMG and W3C? Provide examples.

Question 2 [10 marks]

The programming language C evolved from the programming language B by adding “types”. Furthermore, C++ then introduced the object oriented paradigm and, in doing so, also the notion of user defined ‘abstract data types’. Clarify the difference by providing examples of such “types” [5 marks] and “abstract data types” [5 marks].

Question 3 [10 marks]

One of the key concepts within the context of event driven programming is that of a ‘Dispatcher’. What is the role of the Dispatcher?

Question 4 [10 marks]

Many of today’s software services are provided via a Web API using JSON as a standard. Explain why some of these software services incur a cost when used on a large scale but they are free when used on a small scale.

Question 5 [10 marks]

One approach to achieve interoperability between software systems is to use JSON (JavaScript Object Notation). JSON is an “open standard” that uses human-readable text. Why is it relevant that a standard is “open”? [7 marks]
Provide an example of a software standard that is not “open”. [3 marks]

Question 6 [10 marks]

Consider a software system that controls a recycling machine for returnable bottles, cans and crates. A customer can return all three types of item on the same occasion. The system must check, for each item, what type has been returned. The system will register how many items each customer returns and when the customer asks for a receipt, the system will print out what was deposited, the value of the returned items and the total return sum that will be paid to the customer.

This system has been used as an example during the lecture and the practical sessions. A draft class diagram is provided in Figure 1.

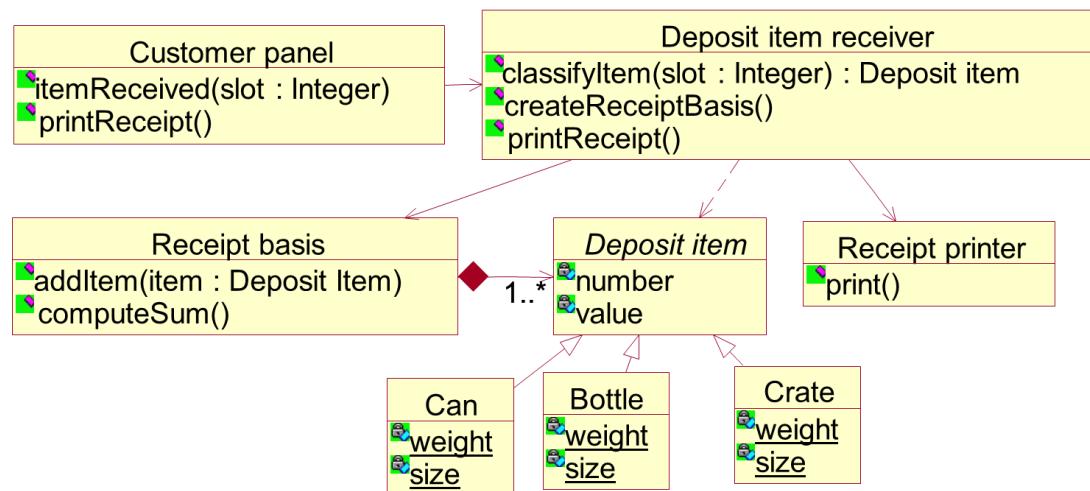


Figure 1 Draft class diagram for Recycling Machine Implementation

Provide a detailed answer to the following question [10 marks]:

- What is the role or responsibility of the 'Receipt Basis' class?

Question 5 [40 marks]

Freemantle Triangle University wants you to develop a web site for an essay writing contest. From an initial meeting with the University's Marketing Team you learn the following: staff and students can register on the web site and upload an essay of up to 800 words about their favourite activities at Freemantle Triangle University; essays that are longer than 800 words should be automatically rejected by the system. It is important for the University that the web site for the contest has the same 'look and feel' as the existing web sites of the University. When all essays are uploaded the public can read these essays online and vote for the best one. The winner's essay will then feature on the main University web site. As the contest should also serve to recruit new students it is important to track where the votes are coming from and what other parts of the University's web site are visited by the same person.

- In this scenario discuss the role that HTML [4 marks], CSS [4 marks], JavaScript [6 marks] and PHP [6 marks] may play.
- With reference to the scenario above, critically discuss in detail how cookies [10 marks], IP addresses [6 marks] and passwords [4 marks] could be used to establish online identity.

END OF EXAM QUESTIONS